

Tutorial

Meteor Scatter: An Overview

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Invited Review Paper

Abstract—Meteor burst communication has drawn a great deal of attention in recent years as the perceived vulnerability of HF and satellite-based communication systems has increased. This relatively unknown media is surveyed, concentrating on recent advances in the state of knowledge.

I. INTRODUCTION

THE FACT that the ionized trails of meteors entering the Earth's atmosphere will reflect radio waves has been known since the early 1930's when Pickard [1] associated unexpected bursts of short-wave radio propagation with the occurrences of meteor showers. It was not until the late 1940's, when technology had pushed the radio spectrum well into the VHF band, that communication engineers became interested in the meteor propagation phenomenon. The late 1940's and 1950's saw the peak of meteor scatter research. In both the United States and Canada, experimental systems were established to analyze this new communication medium [2]–[13] and the December 1957 issue of the PROCEEDINGS OF THE IRE was devoted to this topic.

During the 1950's efforts were undertaken to model and predict the arrival of meteors [14]–[17], characterize meteor echoes, use meteor trails to measure high-altitude winds [11], and model the communication performance of meteor scatter systems [6], [18], [19]. The technology of the time provided limited storage and signal-processing capability [4]–[6], [8] so that communication systems were of limited usefulness. Just as new transistor technology began to advance to the point where more useful systems could be developed [20], satellites became the technology of choice for long-range communication. By the middle to late 1960's, interest in meteor communication had waned. The state of knowledge of meteor communication in the middle 1960's was surveyed in the classic paper by Sugar [21] and the book by McKinnley [22]. During the 1970's meteor scatter research was primarily scientific, using meteor trail echoes as a tool for characterizing the middle atmosphere [23].

In recent years the perceived vulnerability of satellite-based communication systems, coupled with improvements in technology, especially inexpensive microprocessor-based systems, has led to a resurgence of interest in both HF and meteor

scatter as survivable backups to the high-throughput systems. Recognizing the potential value of meteor communication for military applications, Oetting [24] surveyed the state of meteor scatter in terms of its applicability to military missions.

In addition to military applications, meteor communication can be useful in a number of civilian applications, especially remote sensing where it can provide an inexpensive alternative to conventional communication techniques for gathering data from large networks of sensors [25].

Meteor scatter has drawn a great deal of attention because it has several potential advantages over conventional HF for long-haul digital communication applications. The meteor propagation mechanism does not depend upon ionospheric reflection, so a single frequency can be used for the operation of a network, as contrasted to HF which requires both a number of frequency allocations and frequency management to adapt to day-to-day, diurnal, and seasonal variations in the maximum usable frequency of the ionosphere and crowding and interference in many bands. Control of meteor communication networks is, therefore, less complicated.

While the maximum range of single-hop meteor communication (2000 km max) is less than that of HF systems, conventional HF is subject to a skip range below which communication is not possible. Meteor burst is not subject to the skip zone phenomenon and it has been demonstrated [26], [27] that if antenna patterns and link protocols are properly designed, continuous connectivity up to the single-hop maximum range of 2000 km can be maintained. Meteor scatter utilizes higher frequencies than HF (40–100 MHz for meteor scatter versus 2–30 MHz for HF), so it is less subject to absorption and blackout conditions which frequently occur at high latitudes. Finally, because the scattering area of a meteor trail is small compared to that of the HF layer, the ground illumination footprint of a meteor trail is smaller than that of HF so the meteor channel has a certain degree of inherent protection against ground-based interceptors and jammers.

A survey of recent meteor-scatter related literature indicates that research is proceeding in two general directions: improving the understanding of the propagation mechanism and developing improved technology and techniques for meteor scatter communication. Recent work in the propagation area is focusing on improving the basic model for meteor-scatter propagation in the VHF band, including the effect of ionospheric phenomena and propagation mechanisms other than

Manuscript received August 1987.

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IEEE Log Number 8823641.

0018-926X/88/1200-1813\$01.00 © 1988 IEEE

meteor trails. Cannon [28] has investigated the effect of the Faraday rotation mechanism and demonstrated that depending upon the location and distance of the link relative to the magnetic field of the earth, loss due to polarization rotation could be observed. At high latitudes there is increasing interest in meteor scatter as an alternative to HF for long-distance communication. The U.S. Air Force has established a high-latitude test bed to study the meteor scatter propagation mechanism in general and to assess the effects of high latitude on meteor scatter communication in particular. Ostergaard *et al.* [29] describe the high-latitude testbed and present preliminary data. Weitzen [30] describes the software database created from the trails observed on the link and presents examples of the analysis. Sowa *et al.* [31] summarize in detail the data from the first three months of operation of the link. Weitzen *et al.* [32] characterize the multipath and fading profiles of auroral scatter and sporadic-E propagation in addition to meteor trails, extending the work of [8] and [9]. Prykl *et al.* [33] characterize the Doppler properties of auroral scatter and meteor trails using coherent scatter radar. Of great interest is the effect of polar cap absorption on meteor communication [34]. Using data from recent experiments, several improved models have been developed for analyzing the complicated interaction between the spacial arrival pattern of meteor trails and receiver-transmitter antenna patterns [35], [36].

Communication research is focusing on improving the throughput and reliability of meteor scatter, especially at short ranges. Weitzen *et al.* [37] describe a communication technique to improve the throughput by adapting the data rate to the capacity of the channel at an arbitrary bit error rate. If a large bandwidth is available, this technique has the potential to improve the throughput by a factor of 10. This work has been extended by Abel [38] to determine an optimum fixed data rate for each meteor trail. Milstein *et al.* [39] examine coding techniques to improve the throughput performance. Communicating via meteor scatter at short ranges has received a great deal of interest in recent years to fill the gap between long-range HF communication and line-of-sight/troposcatter communication. Nes [26] describes experiments on the SHAPE system in Europe, including an experiment in which the range was varied. Cannon and Reed [40] describe the operation of the research system "BLOSSOM." Weitzen [27] analyzes the requirements, in terms of protocol design and antenna design, for improving the performance of meteor communication at short ranges. Sinnot *et al.* [41] and Thompson *et al.* [42] describe experiments using a buried antenna. The remainder of the minireview provides an overview of the meteor scatter phenomena, including recent advances in the state of knowledge.

II. RADIO PROPAGATION VIA METEOR TRAILS

Meteors and Meteor Trails

Billions of particles enter the Earth's atmosphere each day, ranging in size from milligram micrometeors to large stones. When an incoming meteor reaches the lower E region of the atmosphere, 90–120 km above the earth, friction due to the increased atmospheric density causes the meteors to burn up, or more precisely to evaporate, producing a trail of positive

ions and free electrons. At formation, trails are tubular clouds, typically, 0.5 m in diameter [43] with typical lengths of 15–30 km [44]. Immediately after formation, the trail begins to expand due to ambipolar diffusion, which is the primary mechanism leading to trail decay.

According to Manning [43] and Eshleman [44], meteors as small as 10^{-5} grams will produce a trail which is visible by radio techniques. Radio observable trails occur when the line defining the axis of the tubular meteor trail is tangent to an ellipsoid of revolution with foci at the receiver and transmitter [22]. This point defines the center of a region of constructive interference known as the first or principal Fresnel zone. In addition to the geometric constraint, for the trail to be radio observable there must be significant ionization at the point of tangency.

The point of tangency is a function of the location of the foci (receiver and transmitter). If the receiver moves far enough from the original location so that at the new point of tangency there is not significant ionization, a trail observed at the original location will not be observed at the new location. Depending on the orientation and length of the trail and the location on the ground relative to the original receiver, the distance from a receiver at which the received signal is down by 10 dB can range from 20 to 100 km. It is this phenomenon that leads to the inherent jam and intercept resistance of meteor communication.

Most meteors orbit the sun in the same direction as the Earth and the other planets. The vast majority of meteors occur in random orbits and are referred to as sporadic meteors. Groupings of meteoric particles, believed to be the remnants of comets, occasionally intersect the Earth. When this occurs, a meteor shower results. A number of regularly occurring meteor showers have been catalogued. Although large numbers of meteors can be observed during shower periods, the vast majority of the meteor trails observed on a day-to-day basis are due to sporadic meteors.

Because of the daily rotation of the Earth in its orbit around the Sun, meteors are swept in by the motion of the Earth in the early morning hours (4–6 am). During the early evening hours, about 6 pm, the orientation of the earth is such that meteors must overtake the Earth. This accounts for the strong diurnal variation observed on most meteor links [14]. The arrival of meteors is described by a Poisson process [21] for which the times between meteor arrivals are exponentially distributed and independent from arrival to arrival.

Underdense Propagation Mechanism

The most frequently observed meteoric propagation mechanism is the underdense meteor trail. These trails are formed by micrometeors with masses from about 10^{-5} to 10^{-3} g [21]. Underdense trails are characterized as having a sufficiently low electron density so that an incident wave passes through the trail. Each electron in the trail scatters the incident electromagnetic wave independently of all other electrons in the trail. A commonly used approximation is to model underdense trails as tubular clouds of free electrons with uniform axial and Gaussian radial electron densities. Eshleman [12] solved the diffusion equation to determine the radial electron density of a tubular trail as a function of time (t) and

integrated the signal contribution from each electron over the radial and axial electron densities to yield a closed form solution for the received power from an underdense meteor trail as

$$P_r(t) = \frac{P_T G_T G_R \lambda^3 q^2 r_e^2 \sin^2 \alpha}{16\pi^2 R_T R_R (R_T + R_R) (1 - \cos^2 \beta \sin^2 \phi)} \cdot \exp\left(\frac{8\pi^2 r_0^2}{\lambda^2 \sec^2 \phi}\right) \exp\left(\frac{-32\pi^2 D t}{\lambda^2 \sec^2 \phi}\right) \quad (1)$$

where

- P_T transmitted power,
- G_T, G_R transmit and receive antenna gains as a function of location in the common volume,
- q electron line density in the trail (e/m) assumed uniform along trail,
- D diffusion constant, a function of trail height (m²/s),
- λ wavelength (m),
- r_e classical radius of the electron (2.818×10^{-15} m),
- R_T, R_R distances (m) from transmitter and receiver, respectively, to the center of the first Fresnel zone,
- r_0 initial effective trail radius (m),
- α angle between the incident electric field and the wave normal from the receiver to the trail,
- 2ϕ angle of incidence formed by the vectors $\vec{R}_T$ and $\vec{R}_R$,
- β orientation of the trail in the plane normal to the plane of incidence containing $\vec{R}_T$ and $\vec{R}_R$.

Inherent in the Eshleman model are a number of assumptions including the requirement that the time required to form the trails is negligible relative to the average decay time and Gaussian radial and uniform axial electron densities. Other cases are discussed in a later paper [45]. Underdense trails are characterized by their rapid rise and exponential decay with time. Figs. 1 and 2 show examples of meteor trails observed at 45 MHz on the 1260-km USAF High Latitude Meteor Burst Test Bed [29], [30]. The three upper trails in Fig. 1 show examples of underdense trails with different amplitudes and decay times.

Meteors occur at random heights from 85–120 km above the earth. Due to the height dependence of the diffusion constant D [46], the decay time is a function of the height at which the trail occurs, and recent data from the high latitude meteor burst test bed [30] indicate that the decay constant is an exponentially distributed random variable with average determined by the technique of Brown and Williams [47] as

$$\tau_{ave} = \frac{\lambda^2 \sec^2 \phi}{32\pi^2 D_{ave}} \quad (2)$$

where

$$D_{ave} = 10.0 \{0.067[127 - 17 \log(f)] - 5.6\} \text{ m}^2/\text{s}$$

and f is the frequency in MHz. The observation that the time constant varies from trail to trail is demonstrated in the three upper trails of Fig. 1.

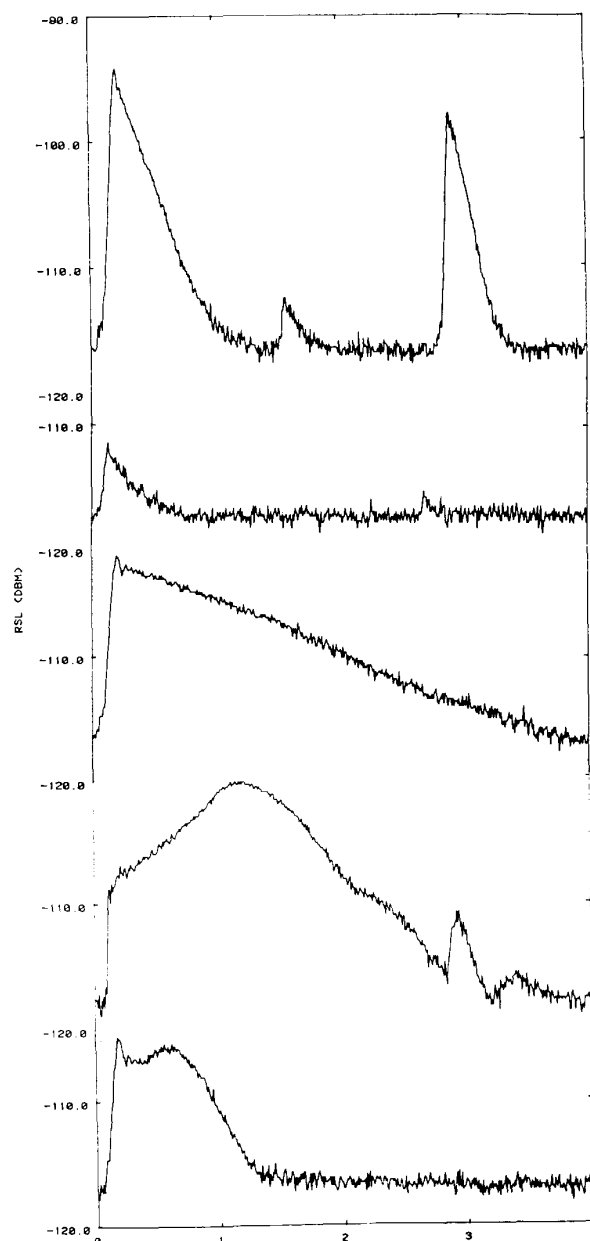


Fig. 1. Examples of meteor trails observed on high latitude test bed [30] at 45 MHz. Top three trails underdense; bottom two trails overdense.

Overdense Meteor Propagation

As the electron density increases, the trail appears less and less like a tubular cloud of independently scattering electron and more and more like a plasma cloud. These trails are referred to as overdense. The model for propagation is much more complicated than for underdense trails due to the complicated interaction between electrons and no exact closed form solution for the return from overdense trails has yet been formulated. Hines and Forsythe [48] assumed that the trails were so dense that there was perfect reflection from the surface of an expanding tubular metallic cylinder with a radius

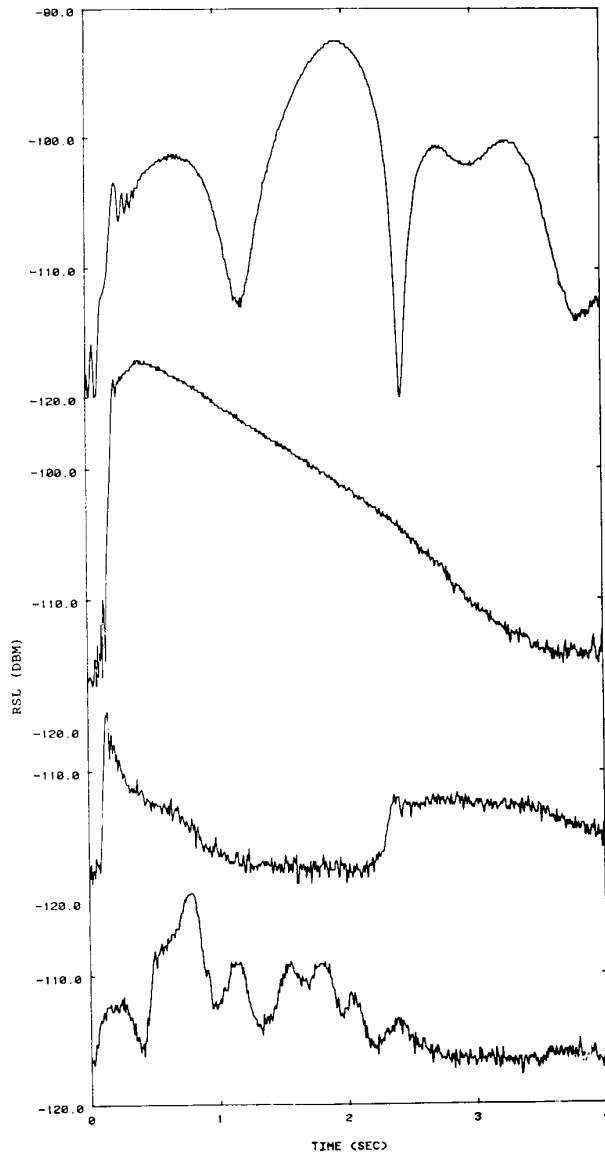


Fig. 2. First trail: large overdense trail with fading; second trail: large trail with characteristics of underdense and overdense trails; third: underdense and overdense trail; fourth: nonspecular meteor trail.

given by the critical radius. The received power due to reflection from an overdense trail approximated by Hines and Forsythe is given by

$$P_r(t) \approx \frac{P_T G_T G_R \lambda^2 \sin^2 \alpha}{32 \pi^2 R_T R_R (R_T + R_R) (1 - \cos^2 \beta \sin^2 \phi)} \cdot \left[\frac{4Dt + r_0^2}{\sec^2 \phi} \ln \left(\frac{r_e q \lambda^2 \sec^2 \phi}{\pi^2 (4Dt + r_0^2)} \right) \right]^{1/2} \quad (3)$$

Overdense trails are characterized by an initial rapid rise as the trail forms, continued slow rise as the metallic cylinder expands, followed by a slow decay as the cylinder diffuses. As the trail continues to diffuse, the density is reduced to the point

where the trail begins to behave like an underdense trail. Overdense trails generally have a longer duration than underdense trails. As the frequency increases, the critical radius of a given trail decreases so that at higher frequencies more trails appear underdense [30]. The lower two trails in Fig. 1 show typical overdense trails at 45 MHz. The first trail in Fig. 2 shows a large overdense trail which is slowly fading due to interference most likely from another meteor. The second trail in Fig. 2 shows a trail which has the slow rise time characteristic of an overdense trail and the exponential decay of an underdense trail.

The transition from underdense to overdense is not sharp; rather, it is gradual. A commonly used approximation for the transition is the electron density at which the return, from a trail at the midpoint with $\beta = 0$, assuming the model of Hines and Forsythe (3) exceeds the return due to (1). Trails with q greater than the transition value are considered overdense, and trails with smaller q are considered underdense. At frequencies in the 50-MHz range, $q = 10^{14}$ e/m is often used as the transition.

It is often erroneously assumed that the received power from overdense trails is always much larger than from underdense trails. From (1) and (3) it is observed that the received power is a function of both the electron density of the trail and the length of the first Fresnel zone, which is a function of trail orientation (β). Depending on the length of the Fresnel zone and the polarization factor α , the return from an underdense trail could be greater than that due to an overdense trail. The third trail in Fig. 2 shows returns from both an underdense and overdense trail within a four second period. The underdense trail return is larger than that from the overdense trail.

Nonspecular Meteor Trails

Occasionally, very large meteor trails form and begin to warp and drift due to high-altitude winds. After several seconds, these trails resemble a cloud rather than a trail. The cloud can drift so that the conditions for specular scatter are satisfied by several points. The result is a trail with little structure and which exhibits fading as shown in the fourth trail in Fig. 2. These trails have been referred to as "nonspecular."

Multipath and Fading Profile of the Meteor Channel

The meteor channel is for the most part a power limited channel as opposed to multipath limited channels such as troposcatter; however, as burst data rates increase [37], it is increasingly important to understand and model multipath and fading on the meteor channel. Several multipath mechanisms have been identified on meteor burst channels. Manning [49] showed that high-altitude wind shears could warp a trail so that portions of the trail initially in the first Fresnel zone, contributing constructively to the signal, would later in time contribute destructively, causing fading. The multipath spread resulting from this mechanism is limited to several Fresnel zones or several hundred nanoseconds at 50 MHz. On occasion, the winds could warp the trail so that secondary areas satisfy the geometric conditions for specular scatter. This mechanism can result in fading as well as delay spreads as

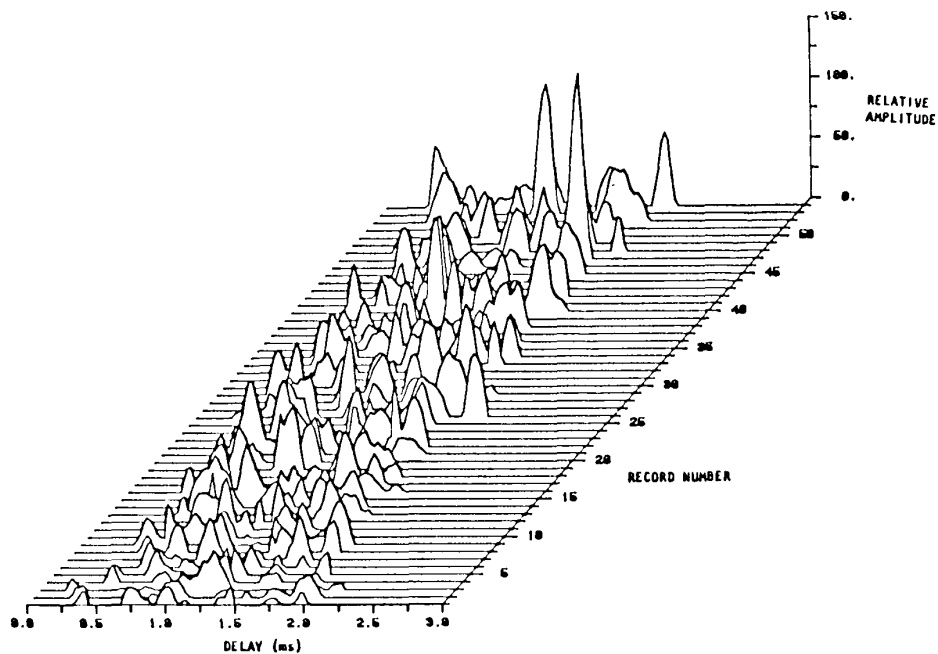


Fig. 3. Series of multipath snapshots showing effect of auroral scatter on meteor communication. Large peaks in frames 47, 53, and 54 are meteor trails.

high as several microseconds for long trails, which has been observed in experiments [8], [9].

The simultaneous occurrence of more than one meteor on a link can cause multipath in excess of 1–2 ms, depending on the geometry. Fortunately, for communication links with 500–1000 W transmitters, this phenomenon is observed on less than one to two percent of trails [8].

Because the scattering mechanism is, for the most part, specular, large Doppler spreads are generally not observed on meteor channels. Reflections from the head of the trail as it forms can cause peak Doppler shifts of 50–75 Hz during the short time that the meteor is transmitting the principal Fresnel zone. Trail drift due to high altitude winds can account for Doppler shifts of typically 2–10 Hz [11, 32].

Other Propagation Mechanisms

No survey of meteor burst communication would be complete without a discussion of propagation mechanisms other than meteor trails which are commonly observed on meteor links. These mechanisms can change the characteristics of the meteor communication channel. Propagation mechanisms, in addition to meteor trails, observed on meteor links include sporadic-E propagation, auroral scatter, ionospheric scatter, diffraction, line-of-sight propagation, and tropospheric scatter.

Sporadic-E

Varieties of sporadic-E occur frequently at all latitudes. When present, sporadic-E reflections exhibit low multipath spreads, slow fading, and high signal levels. The primary multipath mechanism on the sporadic-E channel is, ironically, the occurrence of meteor trails [9], [32]. Typical Doppler spreads are less than 10 Hz [32]. At 40–100 MHz sporadic-E is a low-probability phenomena, which is its main drawback.

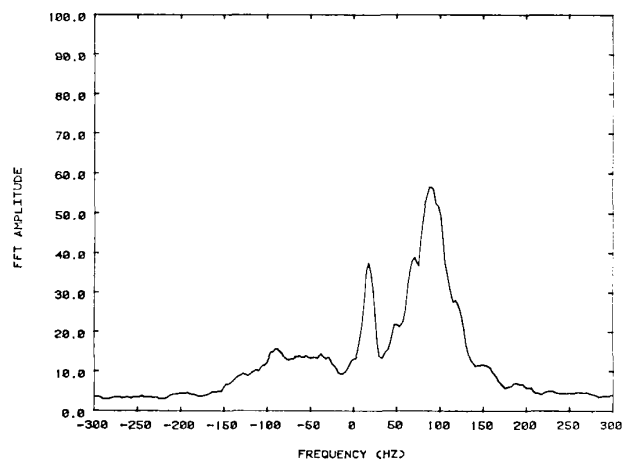


Fig. 4. Doppler profile of combination of auroral scatter and meteor channel. Component at zero Doppler is due to meteor trails, and component centered at approximately 80 Hz is due to auroral scatter.

Auroral Scatter

It is becoming increasingly apparent that at high latitudes auroral scatter can play an important role in determining the performance of meteor burst communication systems. Auroral scatter can transform the relatively benign, though highly intermittent, meteor channel into a relatively high-signal level multipath-limited channel. In a recent experiment on a 640-km link from Anchorage to Bethel in Alaska, rms multipath spreads as high as 1 ms with Doppler spreads in excess of 200 Hz were observed when auroral scatter was the dominant propagation mechanism. Communication in this environment requires changes in the modulation, data rate, and signaling protocols [32]. Figs. 3 and 4 show the multipath and fading profile, respectively, of a VHF (42 MHz) channel during a

large auroral event during July 1985. In Fig. 3 the large peaks correspond to meteor trails and the background represents the auroral scatter/component.

Diffraction, Troposcatter, and Line of Sight

At ranges less than several hundred kilometers, meteor scatter may not be the dominant propagation mechanism. There is an increasing appreciation of the roles of diffraction, tropospheric scatter, and line-of-sight propagation on meteor scatter communications [26], [27].

III. SUMMARY

Improvements in technology and the requirement for reliable low data rate communication have rejuvenated interest in meteor scatter. A number of on-going research efforts indicate that with the improved technology meteor scatter can satisfy a number of existing communication requirements. In the next few years a number of meteor communication systems will become operational, many incorporating the latest technological innovations.

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